

# Mahmad Shahabaz

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## PROFESSIONAL SUMMARY

Motivated Computer Science undergraduate with hands-on experience designing and deploying secure, AI-augmented systems. Proficient in full-stack development, backend engineering, and applied machine learning. Proven ability to deliver end-to-end solutions—from embedded hardware interfaces to cloud-deployable REST services—with a strong focus on security, scalability, and real-world impact. Currently pursuing a B.E. in Computer Science Engineering at BMS College of Engineering, Bengaluru.

## TECHNICAL SKILLS

|                    |                                                                                          |
|--------------------|------------------------------------------------------------------------------------------|
| Languages          | C, Java, Python, JavaScript, Kotlin                                                      |
| Frontend           | React.js, Next.js, HTML5, CSS3, React Router, Single Page Applications (SPA)             |
| Backend & APIs     | Node.js, Express.js, Spring Boot, Flask, RESTful APIs, JWT Authentication                |
| Databases          | MySQL, MongoDB                                                                           |
| AI / ML / CV       | YOLOv5, OpenCV, Google ML Kit (OCR), Face Recognition                                    |
| Mobile Development | Kotlin (Android), ZXing Barcode SDK                                                      |
| Core CS            | Data Structures & Algorithms, OOP, DBMS, Operating Systems                               |
| Tools & DevOps     | Git, GitHub, VS Code, Linux, Postman, UML                                                |
| Soft Skills        | Problem Solving, Team Collaboration, Time Management, Project Management, Rapid Learning |

## PROFESSIONAL EXPERIENCE

**Software Engineering Intern Government Polytechnic Tumkur**, Tumkur, Karnataka, IN Jan 2024 – May 2025

- Designed and delivered three full-cycle engineering projects covering AI-integrated security systems, full-stack e-learning platforms, and enterprise-grade inventory management — demonstrating end-to-end software development proficiency.
- Collaborated with faculty supervisors and cross-functional student teams using Agile methodologies to meet defined project milestones and deliverables on schedule.
- Authored technical documentation and system architecture diagrams for all developed projects, facilitating knowledge transfer and maintainability.

**Community Outreach Volunteer InCap Group**, Tumkur, Karnataka, IN Jun 2024 – May 2025

- Coordinated community outreach initiatives and events for 50+ participants, improving overall volunteer engagement by streamlining onboarding and communication workflows.
- Designed and produced marketing collateral for campaigns, directly contributing to a measurable increase in volunteer recruitment and community program visibility.
- Mentored and trained new volunteer cohorts, reducing onboarding time through structured orientation sessions and resource preparation.
- Managed stakeholder communications across diverse audiences, building cross-functional relationships to advance organizational goals.

## PROJECTS

**Secure Multi-Bank ATM System** | Python, Flask, MySQL, YOLOv5, OpenCV, Face Recognition, Telegram API 2024 – 2025

- Architected a unified multi-bank ATM platform using a single RFID-based affinity card, eliminating the need for multiple bank-specific cards and improving user convenience.
- Implemented robust multi-factor authentication (MFA) combining face recognition (DeepFace/OpenCV) and OTP verification,

- reducing unauthorized access risk to near zero.
- Integrated an AI-driven real-time threat detection module using YOLOv5 + OpenCV that classifies suspicious activities at ATM terminals and dispatches automated Telegram alerts to security personnel within seconds.
- Engineered an emergency reverse-PIN feature enabling covert distress signaling without alerting potential perpetrators, enhancing customer safety.
- Built a secure, scalable backend with Python (Flask) and MySQL, employing parameterized queries and session management to prevent SQL injection and session hijacking.

**LoopLearn — AI-Powered Online Learning Platform** | React.js, Node.js, Express.js, MongoDB, JWT, 2024 – 2025  
REST APIs

- Engineered a fully responsive full-stack e-learning platform supporting course browsing, lecture streaming, and student/instructor enrollment workflows across web and mobile viewports.
- Designed and implemented 15+ RESTful API endpoints for user authentication, course management, enrollment, and role-based content delivery with JWT-secured routes.
- Implemented role-based access control (RBAC) differentiating student and instructor permissions, protecting sensitive course content and administrative functions.
- Integrated an AI-powered learning assistant and intelligent search module, enabling context-aware content recommendations and reducing student query resolution time.
- Architected a Single Page Application (SPA) using React Router with lazy-loading and component-level code splitting, achieving faster initial load times and smoother navigation.
- Optimized MongoDB schema design for course, enrollment, and user data, ensuring sub-100ms query latency under expected concurrent load.

**PIMS — Product Inventory & Expiry Management System** | Kotlin, Spring Boot, MySQL, ZXing, 2024 – 2025  
Google ML Kit, JWT

- Developed an Android-based inventory management application featuring barcode scanning, OCR-based product recognition, expiry tracking, billing, and supplier management.
- Integrated ZXing Barcode Scanner and Google ML Kit OCR to automate product data capture, reducing manual entry and improving inventory accuracy.
- Built an automated expiry monitoring and low-stock alert system to minimize inventory waste and prevent stock shortages.
- Developed billing, invoice generation, and supplier management modules to streamline retail operations and procurement workflows.
- Implemented a billing and invoice generation module supporting multi-product transactions, tax calculation, and professional PDF invoice output with full sales record history.
- Designed a secure and scalable Spring Boot backend with MySQL and JWT authentication, supporting real-time inventory updates and multi-user access.

EDUCATION

**Bachelor of Engineering (B.E.) — Computer Science & Engineering** 2025 – 2028 (Expected)  
BMS College of Engineering, Bengaluru, India — CGPA: 7.23

- Capstone Project: *Secure Multi-Bank ATM System Backend* — AI-integrated security and authentication platform.
- Active member of the **Rotaract Club** (2025–2028) and **NSS (National Service Scheme)** — contributing to community service and leadership development.

**Diploma — Computer Science & Engineering** 2022 – 2025  
Government Polytechnic Tumkur, Tumkur, India —

CERTIFICATIONS & ACHIEVEMENTS

- Full Stack Development** — Infosys Springboard 2024
- Git Foundation Certification** — Infosys (TechA Program) 2026
- Mastering Git Certificate** — Infosys 2026
- JobReady Employability Skills** — Wadhwani Foundation 2023
- Google Student Ambassador Program — Pitch Night Participant**; presented an original innovation project to a panel of judges, demonstrating entrepreneurial thinking and technical communication skills. 2026

LANGUAGES

English (Professional Proficiency)   Hindi (Native)   Kannada (Native)   Urdu (Conversational)